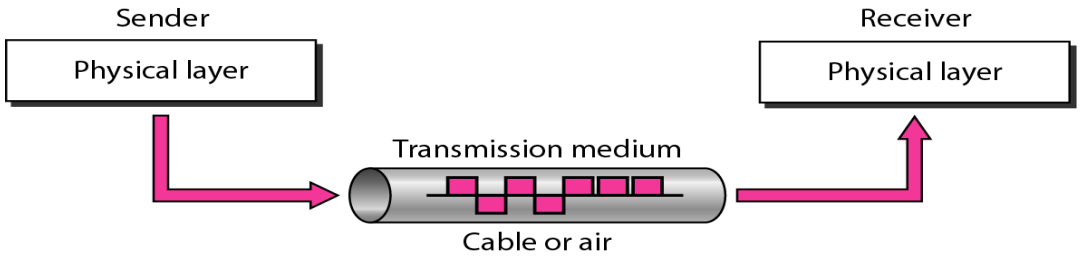
UNIT 3

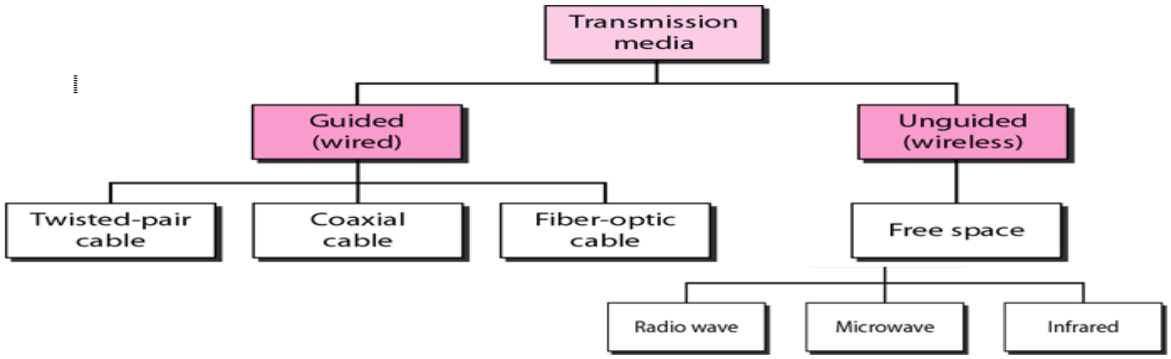
TRANSMISSION MEDIA

- A transmission media define as anything that can carry information from a source to a destination.

Transmission medium and physical layer



Classes of transmission media



GUIDED MEDIA

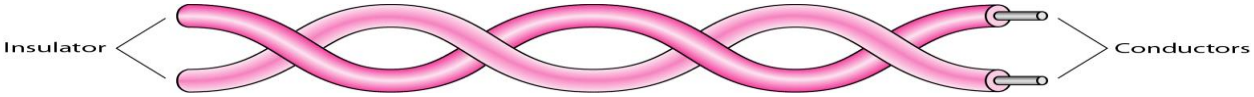
- It is defined as the physical medium through which the signals are transmitted.
- It is also known as Bounded media.

Types of Guided media:

- ❑ Twisted-pair cable
- ❑ Coaxial cable
- ❑ Fiber-optic cable.

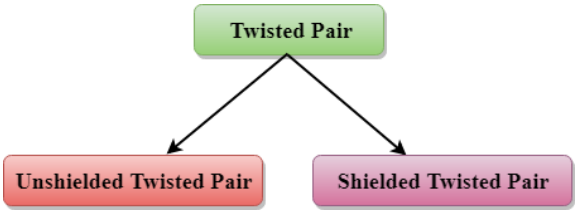
Twisted- pair cable :

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair consists of two insulated copper wires, twisted together, as shown in Figure.



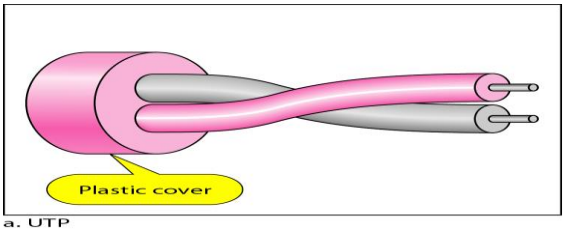
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- In addition to the signal , the wire carries interference (noise) and crosstalk
- It may affect both wires and create unwanted signals.
- If the two wires are parallel, the effect of these unwanted signals is reduced because they are at different locations relative to the noise or crosstalk sources . This results in a difference at the receiver. By twisting the pairs, a balance is maintained.

Types of Twisted pair:



Unshielded Twisted Pair:

An unshielded twisted pair is widely used in telecommunication. Following are the categories of the unshielded twisted pair cable:



a. UTP

- **Category 1:** Category 1 is used for telephone lines that have low-speed data.
- **Category 2:** It can support upto 4Mbps.
- **Category 3:** It can support upto 16Mbps.
- **Category 4:** It can support upto 20Mbps. Therefore, it can be used for long-distance communication.
- **Category 5:** It can support upto 200Mbps.

Advantages Of Unshielded Twisted Pair:

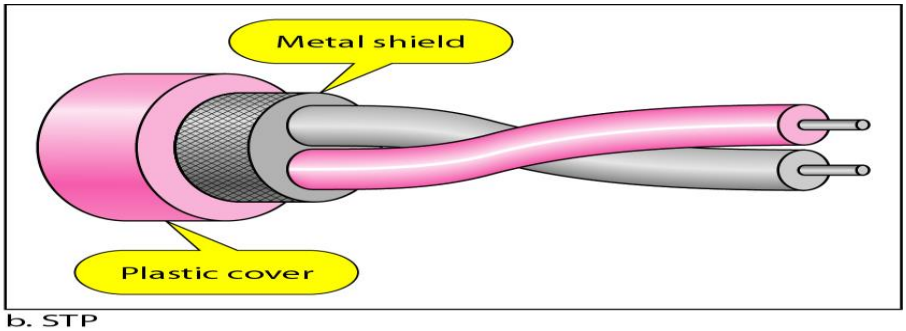
- It is cheap.
- Installation of the unshielded twisted pair is easy.
- It can be used for high-speed LAN.

Disadvantage:

- This cable can only be used for shorter distances because of attenuation.

Shielded Twisted Pair

- A shielded twisted pair is a cable that contains the mesh surrounding the wire that allows the higher transmission rate.



b. STP

Characteristics of Shielded Twisted Pair:

- The cost of the shielded twisted pair cable is not very high and not very low.
- An installation of STP is easy.
- It has higher capacity as compared to unshielded twisted pair cable.
- It has a higher attenuation.
- It is shielded that provides the higher data transmission rate.

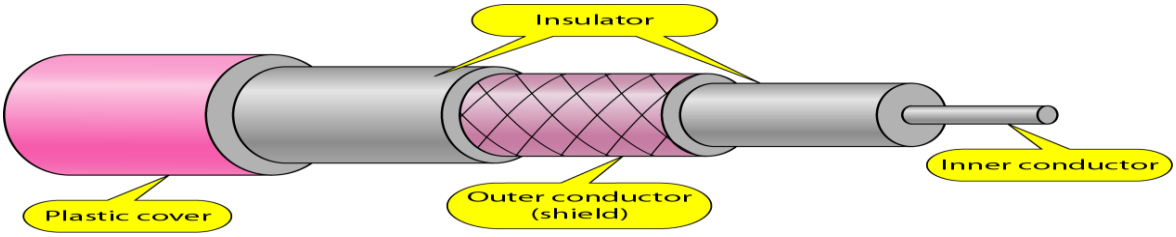
Disadvantages

- It is more expensive as compared to UTP and coaxial cable.
- It has a higher attenuation rate.

Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- It has a higher frequency as compared to Twisted pair cable.

- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).



Coaxial cable is of two types:

- Baseband transmission: It is defined as the process of transmitting a single signal at high speed.
- Broadband transmission: It is defined as the process of transmitting multiple signals simultaneously.

Advantages Of Coaxial cable:

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages Of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

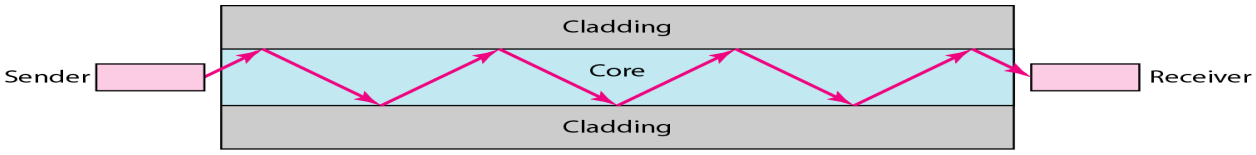
Fibre Optic / Optiacl Fibre

- Fibre optic cable is a cable that uses electrical signals for communication.
- Optical fibre is coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:



Basic elements of Fibre optic cable:



- Core: The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre.
- Cladding: The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.
- Jacket: The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

Advantages of fibre optic cable over copper:

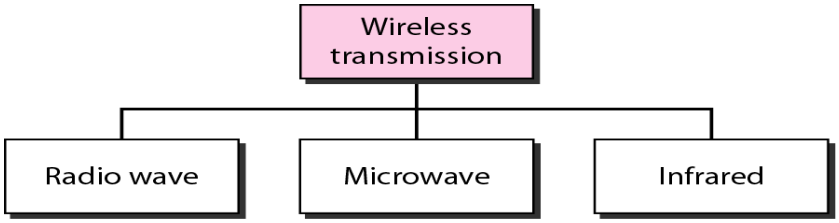
- **Greater Bandwidth**: The fibre optic cable provides more bandwidth as compared copper. Therefore, the fibre optic carries more data as compared to copper cable.
- **Faster speed**: Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.
- **Longer distances**: The fibre optic cable carries the data at a longer distance as compared to copper cable.
- **Better reliability**: The fibre optic cable is more reliable than the copper cable as it is immune to any temperature changes while it can cause obstruct in the connectivity of copper cable.
- **Thinner and Sturdier**: Fibre optic cable is thinner and lighter in weight so it can withstand more pull pressure than copper cable.

Disadvantages There are some disadvantages in the use of optical fiber.

- **Very hard Installation and maintenance**. Fiber-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.
- **Unidirectional light propagation**. Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.
- **High Cost**. The cable and the interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fiber cannot be justified.

UNGUIDED MEDIA: WIRELESS

Unguided media transport electromagnetic waves without using a physical conductor. This type of communication is known as wireless communication



Unguided signal can travel from the source to destination in several ways:

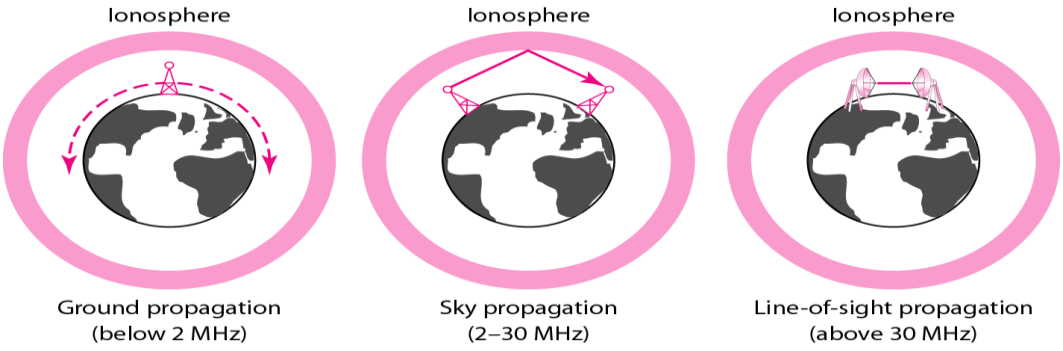
1.Ground Propagation:

- Radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- The low frequency signal follow the curvature of the planet.
- Distance depends on the amount of the power.

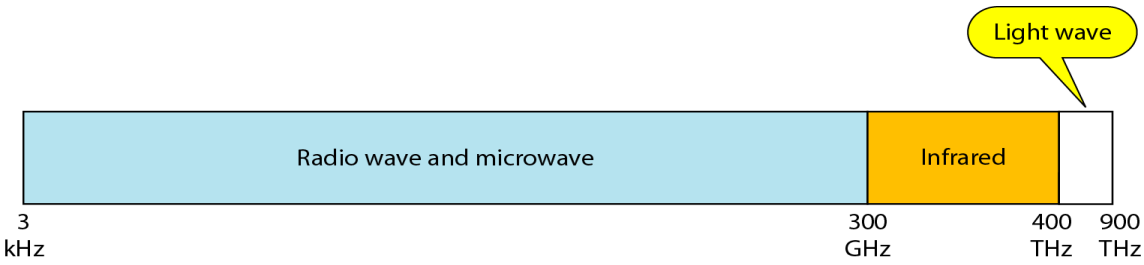
2.Sky Propagation:

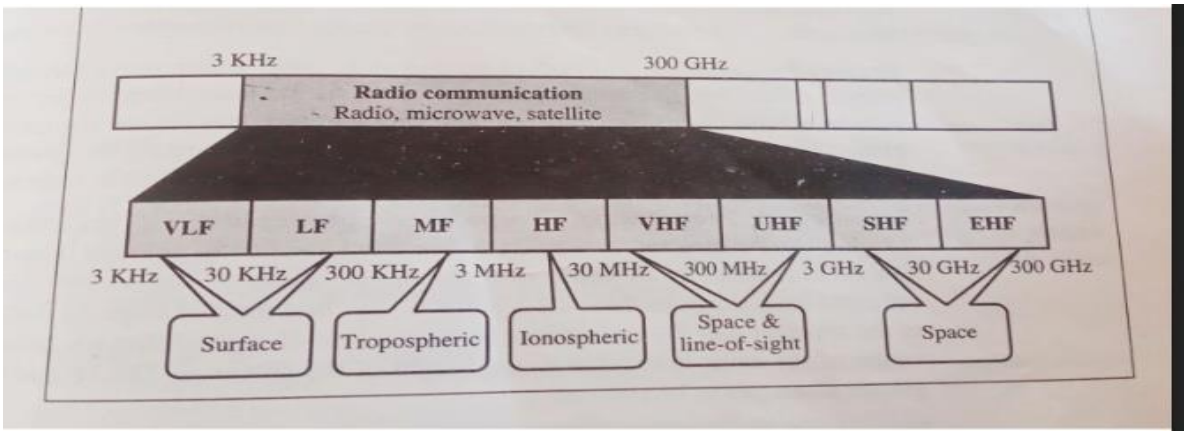
- Higher frequency radio radiate upward into the ionosphere where they are reflected back to the earth.
- Sky propagation allow for greater distance with lower power output.

3.line-of-sight Propagation: Very high frequency signals are transmitted in straight lines directly from antenna to antenna.



Radio Waves : An electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges, called bands, each regulated by government authorities.





Band	*Range*	*Propagation*	*Application*
VLF (very low frequency)	3–30 kHz	Ground	Long-range radio navigation
LF (low frequency)	30–300 kHz	Ground	Radio beacons and navigational locators
MF (middle frequency)	300 kHz–3 MHz	Sky	AM radio
HF (high frequency)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft communication
VHF (very high frequency)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
UHF (ultrahigh frequency)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
SHF (superhigh frequency)	3–30 GHz	Line-of-sight	Satellite communication
EHF (extremely high frequency)	30–300 GHz	Line-of-sight	Radar, satellite

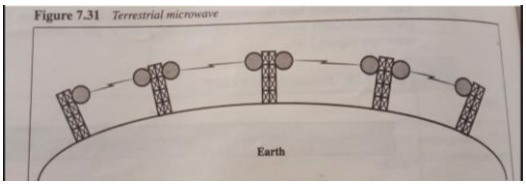
Characteristics :

- It is Omni-Directional
- Between 3 KHz – 1 GHz.
- Radio waves use omnidirectional antenna.
- Radio waves used for multicast communication, such as radio and television.
- Sky Propagation. This makes radio waves a good candidate for long-distance broadcasting such as AM radio.

Disadvantage:

- The radio waves transmitted by one antenna are susceptible to interference by another antenna .

Microwave Transmission:

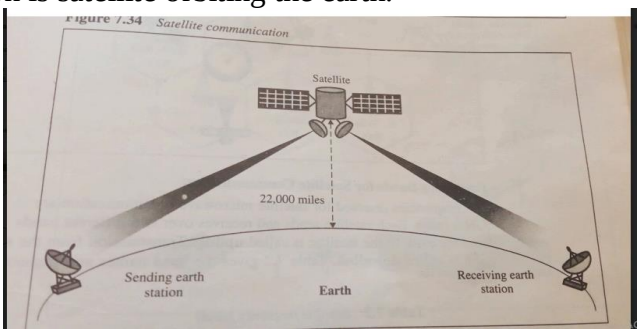


The following describes some characteristics of microwave propagation:

1. Microwave propagation is line-of-sight.
2. Towers use the mounted antennas to send signal and are far apart need to be very tall.
3. Repeaters are often needed for long distance communication.
4. Very high-frequency microwaves cannot penetrate walls.
5. ***Unidirectional*** Antenna Microwaves need unidirectional antennas that send out signals in one direction. Two types of antennas are used for microwave communications: the **parabolic** dish and the **horn**.

Satellite Transmission :

- It is like a line-of-sight microwave transmission.
- One of the station is satellite orbiting the earth.



Advantages:

- High quality communication.
- It allow microwave signals to span continents and oceans with a single bounce

Disadvantages :

- Extremely expensive.
- Use of certain portions of the band requires permission from authorities.

Infrared waves :

- It ranges frequencies from 300 GHz to 400 THz (wavelengths from 1 mm to 770 nm).
- It can be used for short-range communication.
- Infrared waves, having high frequencies, cannot penetrate walls.
- A short-range communication system in one room cannot be affected by another system in the next room. Ex : remote control.
- Line-of-sight propagation.

Disadvantages :

- Infrared signals useless for long-range communication.
- We cannot use infrared waves outside a building because the sun's rays contain infrared waves that can interfere with the communication.